2.4.6 Functions of Operators

Let $F(\hat{A})$ be a function of an operator $\hat{A}$. If $\hat{A}$ is a linear operator, we can Taylor expand $F(\hat{A})$ in a power series of $\hat{A}$:

$$F(\hat{A}) = \sum_{n=0}^{\infty} a_n \hat{A}^n, \tag{2.108}$$

where a_n is just an expansion coefficient. As an illustration of an operator function, consider $e^{a\hat{A}}$, where a is a scalar which can be complex or real. We can expand it as follows:

$$e^{a\hat{A}} = \sum_{n=0}^{\infty} \frac{a^n}{n!} \hat{A}^n = \hat{I} + a\hat{A} + \frac{a^2}{2!}\hat{A}^2 + \frac{a^3}{3!}\hat{A}^3 + \cdots. \tag{2.109}$$

Commutators involving function operators
If $\hat{A}$ commutes with another operator $\hat{B}$, then $\hat{B}$ commutes with any operator function that depends on $\hat{A}$:

$$[\hat{A},\ \hat{B}] = 0 \quad \Longrightarrow \quad [\hat{B},\ F(\hat{A})] = 0; \tag{2.110}$$

in particular, $F(\hat{A})$ commutes with $\hat{A}$ and with any other function, $G(\hat{A})$, of $\hat{A}$:

$$[\hat{A},\ F(\hat{A})] = 0, \qquad [\hat{A}^n,\ F(\hat{A})] = 0, \qquad [F(\hat{A}),\ G(\hat{A})] = 0. \tag{2.111}$$

Hermitian adjoint of function operators
The adjoint of $F(\hat{A})$ is given by

$$[F(\hat{A})]^\dagger = F^*(\hat{A}^\dagger). \tag{2.112}$$

Note that if $\hat{A}$ is Hermitian, $F(\hat{A})$ is not necessarily Hermitian; $F(\hat{A})$ will be Hermitian only if F is a real function and $\hat{A}$ is Hermitian. An example is

$$(e^{\hat{A}})^\dagger = e^{\hat{A}^\dagger}, \qquad (e^{i\hat{A}})^\dagger = e^{-i\hat{A}^\dagger}, \qquad (e^{i\alpha\hat{A}})^\dagger = e^{-i\alpha^*\hat{A}^\dagger}, \tag{2.113}$$

where α is a complex number. So if $\hat{A}$ is Hermitian, an operator function which can be expanded as $F(\hat{A}) = \sum_{n=0}^{\infty} a_n \hat{A}^n$ will be Hermitian only if the expansion coefficients a_n are real numbers. But in general, $F(\hat{A})$ is not Hermitian even if $\hat{A}$ is Hermitian, since

$$F^*(\hat{A}^\dagger) = \sum_{n=0}^{\infty} a_n^* (\hat{A}^\dagger)^n. \tag{2.114}$$

Relations involving function operators
Note that

$$[\hat{A},\ \hat{B}] \neq 0 \quad \Longrightarrow \quad [\hat{B},\ F(\hat{A})] \neq 0; \tag{2.115}$$

in particular, $e^{\hat{A}}e^{\hat{B}} \neq e^{\hat{A}+\hat{B}}$. Using (2.109) we can ascertain that

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}e^{[\hat{A},\ \hat{B}]/2}, \tag{2.116}$$

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A},\ \hat{B}] + \frac{1}{2!}[\hat{A},\ [\hat{A},\ \hat{B}]] + \frac{1}{3!}[\hat{A},\ [\hat{A},\ [\hat{A},\ \hat{B}]]] + \cdots. \tag{2.117}$$

2.4.7 Inverse and Unitary Operators

Inverse of an operator: Assuming it exists[3] the *inverse* $\hat{A}^{-1}$ of a linear operator $\hat{A}$ is defined by the relation

$$\hat{A}^{-1}\hat{A} = \hat{A}\hat{A}^{-1} = \hat{I}, \tag{2.118}$$

where $\hat{I}$ is the unit operator, the operator that leaves any state $|\psi\rangle$ unchanged.

Quotient of two operators: Dividing an operator $\hat{A}$ by another operator $\hat{B}$ (provided that the inverse $\hat{B}^{-1}$ exists) is equivalent to multiplying $\hat{A}$ by $\hat{B}^{-1}$:

$$\frac{\hat{A}}{\hat{B}} = \hat{A}\hat{B}^{-1}. \tag{2.119}$$

The side on which the quotient is taken matters:

$$\frac{\hat{A}}{\hat{B}} = \hat{A}\frac{\hat{I}}{\hat{B}} = \hat{A}\hat{B}^{-1} \quad \text{and} \quad \frac{\hat{I}}{\hat{B}}\hat{A} = \hat{B}^{-1}\hat{A}. \tag{2.120}$$

In general, we have $\hat{A}\hat{B}^{-1} \neq \hat{B}^{-1}\hat{A}$. For an illustration of these ideas, see Problem 2.12. We may mention here the following properties about the inverse of operators:

$$\left(\hat{A}\hat{B}\hat{C}\hat{D}\right)^{-1} = \hat{D}^{-1}\hat{C}^{-1}\hat{B}^{-1}\hat{A}^{-1}, \qquad \left(\hat{A}^{n}\right)^{-1} = \left(\hat{A}^{-1}\right)^{n}. \tag{2.121}$$

Unitary operators: A linear operator $\hat{U}$ is said to be *unitary* if its inverse $\hat{U}^{-1}$ is equal to its adjoint $\hat{U}^{\dagger}$:

$$\hat{U}^{\dagger} = \hat{U}^{-1} \qquad \text{or} \qquad \hat{U}\hat{U}^{\dagger} = \hat{U}^{\dagger}\hat{U} = \hat{I}. \tag{2.122}$$

The product of two unitary operators is also unitary, since

$$(\hat{U}\hat{V})(\hat{U}\hat{V})^{\dagger} = (\hat{U}\hat{V})(\hat{V}^{\dagger}\hat{U}^{\dagger}) = \hat{U}(\hat{V}\hat{V}^{\dagger})\hat{U}^{\dagger} = \hat{U}\hat{U}^{\dagger} = \hat{I}, \tag{2.123}$$

or $(\hat{U}\hat{V})^{\dagger} = (\hat{U}\hat{V})^{-1}$. This result can be generalized to any number of operators; the product of a number of unitary operators is also unitary, since

$$\begin{aligned}(\hat{A}\hat{B}\hat{C}\hat{D}\cdots)(\hat{A}\hat{B}\hat{C}\hat{D}\cdots)^{\dagger} &= \hat{A}\hat{B}\hat{C}\hat{D}(\cdots)\hat{D}^{\dagger}\hat{C}^{\dagger}\hat{B}^{\dagger}\hat{A}^{\dagger} = \hat{A}\hat{B}\hat{C}(\hat{D}\hat{D}^{\dagger})\hat{C}^{\dagger}\hat{B}^{\dagger}\hat{A}^{\dagger} \\ &= \hat{A}\hat{B}(\hat{C}\hat{C}^{\dagger})\hat{B}^{\dagger}\hat{A}^{\dagger} = \hat{A}(\hat{B}\hat{B}^{\dagger})\hat{A}^{\dagger} \\ &= \hat{A}\hat{A}^{\dagger} = \hat{I}, \end{aligned} \tag{2.124}$$

or $(\hat{A}\hat{B}\hat{C}\hat{D}\cdots)^{\dagger} = (\hat{A}\hat{B}\hat{C}\hat{D}\cdots)^{-1}$.

Example 2.10 (Unitary operator)
What conditions must the parameter ε and the operator $\hat{G}$ satisfy so that the operator $\hat{U} = e^{i\varepsilon\hat{G}}$ is unitary?

[3] Not every operator has an inverse, just as in the case of matrices. The inverse of a matrix exists only when its determinant is nonzero.